

Wenbo Deng

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EDUCATION

The Chinese University of Hong Kong, Shenzhen

Computer and Information Engineering, Master of Philosophy

Sep. 2024 - Present

➤ GPA: 3.7/4.0

➤ TOEFL: 101

Shenzhen, China

Hefei University of Technology (Project 211 University)

Applied Physics, Bachelor

Sep. 2020 – Jun. 2024

➤ GPA: 3.81/4.3 (Rank: 5/68)

Hefei, China

RESEARCH

G²RPPO-A: Guided Group Relative Policy Optimization with Adaptive Guidance

Yongxin Guo*, **Wenbo Deng***, Zhenglin Cheng, Xiaoying Tang *Mar. 2025 – Aug. 2025*

Advisor: Dr. Yongxin Guo, Alibaba Group; Prof. Xiaoying Tang, The Chinese University of Hong Kong, Shenzhen

- Worked as the team leader.
- Proposed the guidance mechanism for GRPO algorithm.
- Explored the influence of guided ratio in the Guided GRPO mechanism.
- Implemented adaptive guidance length based on the model performance during training.
- Introduced Curriculum Learning into our training data, which further improved the training effectiveness.

RESEARCH EXPERIENCE

Mixture of attention

Oct. 2024 – Mar. 2025

Advisor: Dr. Yongxin Guo, Alibaba Group; Prof. Xiaoying Tang, The Chinese University of Hong Kong, Shenzhen

- Worked as the team leader.
- Tried to introduce MoE framework into attention mechanism.
- Researched the attention matrices of each head of each layer in ViT during its inference on the CIFAIR-10 dataset.
- Pre-set a threshold which decides how much attention can be regarded as low. For elements that have low attention score, rough calculation is implemented.
- Exploit CUDA kernel programming to implement our newly proposed sparse multiplication.

PROJECT

Code-learning-on-attention | Contributor

- Theoretical study of Transformer structure from the programming aspect.
- Readers can better master the technical details in Transformer.

Fine-tune-ViT | Contributor

- A beginner-friendly guide to fine-tuning has been compiled, simplifying its core concepts from an original perspective.
- Worked fine on the explanation of the concept of fine-tune for my own junior lab mates.

AWARD

Third-class scholarship of Hefei University of Technology	<i>12/2020-12/2021</i>
Third-class scholarship of Hefei University of Technology	<i>12/2021-12/2022</i>
Second-class scholarship of Hefei University of Technology	<i>12/2022-12/2023</i>
Honor of Triple-A Student of Hefei University of Technology	<i>12/2021-12/2022</i>

INTERNSHIP

Sichuan Energy Investment Jinding Industry&Finance Holding Group Co.Ltd.

Department: Department of Information Technology *Jul. 2022 – Aug. 2022*

- Investigated the existing interference server and analyzed the advantages and disadvantages of each system.
- Contributed to the daily operations, maintenance, and management of employee information.